

数据备份与恢复方案

文档版本：v1.0

编写日期：2026 年 3 月 14 日

开发负责人：王舟

一、备份策略

1.1 备份范围

数据类型	备份内容	备份频率	保留周期
数据库	legal_minds.db	每日	30 天
文件存储	storage/files/	每日增量	30 天
配置文件	.env, config/	每周	90 天
日志文件	logs/	每周	90 天

1.2 备份类型

类型	说明	适用场景
完全备份	备份全部数据	每周一次
增量备份	仅备份变更数据	每日一次
差异备份	备份自上次完全备份后的变更	可选

1.3 备份时间

每日备份：凌晨 02:00

每周完全备份：周日 01:00

备份保留：30 天

备份存储：本地 + 云端

二、数据库备份

2.1 SQLite 备份脚本

```
#!/bin/bash
# backup_database.sh
```

```

BACKUP_DIR="/backup/legal-minds/database"
DATE=$(date +%Y%m%d_%H%M%S)
DB_PATH="/opt/legal-minds/data/legal_minds.db"

# 创建备份目录
mkdir -p ${BACKUP_DIR}

# 备份数据库
cp ${DB_PATH} ${BACKUP_DIR}/legal_minds_${DATE}.db

# 压缩备份
cd ${BACKUP_DIR}
tar -czf legal_minds_${DATE}.tar.gz legal_minds_${DATE}.db
rm legal_minds_${DATE}.db

# 删除 30 天前的备份
find ${BACKUP_DIR} -name "*.tar.gz" -mtime +30 -delete

# 记录备份日志
echo "[$(date)] 数据库备份完成: legal_minds_${DATE}.tar.gz" >> /var/log/legal

# 计算备份大小
BACKUP_SIZE=$(du -h ${BACKUP_DIR}/legal_minds_${DATE}.tar.gz | cut -f1)
echo "备份大小: ${BACKUP_SIZE}"

```

2.2 WAL 模式备份

```

#!/bin/bash
# backup_wal.sh

DB_PATH="/opt/legal-minds/data/legal_minds.db"
BACKUP_DIR="/backup/legal-minds/database"
DATE=$(date +%Y%m%d_%H%M%S)

# 检查 WAL 文件
if [ -f "${DB_PATH}-wal" ]; then
    # 检查点操作, 将 WAL 写入主数据库
    sqlite3 ${DB_PATH} "PRAGMA wal_checkpoint(TRUNCATE);"
fi

```

```
# 备份数据库
```

```
cp ${DB_PATH} ${BACKUP_DIR}/legal_minds_${DATE}.db
```

```
# 压缩
```

```
tar -czf ${BACKUP_DIR}/legal_minds_${DATE}.tar.gz -C ${BACKUP_DIR} legal_m
```

```
rm ${BACKUP_DIR}/legal_minds_${DATE}.db
```

2.3 备份验证

```
#!/bin/bash
```

```
# verify_backup.sh
```

```
BACKUP_FILE=$1
```

```
if [ -z "${BACKUP_FILE}" ]; then
```

```
    echo "用法: verify_backup.sh <备份文件>"
```

```
    exit 1
```

```
fi
```

```
# 解压备份
```

```
TEMP_DIR=$(mktemp -d)
```

```
tar -xzf ${BACKUP_FILE} -C ${TEMP_DIR}
```

```
# 验证数据库完整性
```

```
DB_FILE=$(find ${TEMP_DIR} -name "*.db" | head -1)
```

```
if [ -z "${DB_FILE}" ]; then
```

```
    echo "❌ 备份文件中未找到数据库文件"
```

```
    exit 1
```

```
fi
```

```
# 检查数据库
```

```
sqlite3 ${DB_FILE} "PRAGMA integrity_check;"
```

```
RESULT=$?
```

```
# 清理
```

```
rm -rf ${TEMP_DIR}
```

```
if [ ${RESULT} -eq 0 ]; then
```

```
    echo "✅ 备份验证通过"
```

```
    exit 0
```

```
else
    echo "❌ 备份验证失败"
    exit 1
fi
```

三、文件存储备份

3.1 增量备份脚本

```
#!/bin/bash
# backup_files.sh

SOURCE_DIR="/opt/legal-minds/storage"
BACKUP_DIR="/backup/legal-minds/files"
DATE=$(date +%Y%m%d_%H%M%S)
LAST_BACKUP=$(ls -t ${BACKUP_DIR}/files_*.tar.gz 2>/dev/null | head -1)

mkdir -p ${BACKUP_DIR}

if [ -n "${LAST_BACKUP}" ]; then
    # 增量备份
    LAST_BACKUP_DATE=$(basename ${LAST_BACKUP} | sed 's/files_//' | sed 's/.tar.gz//')
    echo "上次备份: ${LAST_BACKUP_DATE}"

    tar -czf ${BACKUP_DIR}/files_${DATE}.tar.gz \
        --newer-mtime="${LAST_BACKUP_DATE}" \
        -C $(dirname ${SOURCE_DIR}) \
        $(basename ${SOURCE_DIR})
else
    # 完全备份
    tar -czf ${BACKUP_DIR}/files_${DATE}.tar.gz \
        -C $(dirname ${SOURCE_DIR}) \
        $(basename ${SOURCE_DIR})
fi

# 删除 30 天前的备份
find ${BACKUP_DIR} -name "files_*.tar.gz" -mtime +30 -delete

echo "[$(date)] 文件备份完成: files_${DATE}.tar.gz" >> /var/log/legal-minds-
```

3.2 云同步备份

```
#!/bin/bash
# sync_to_cloud.sh

LOCAL_BACKUP_DIR="/backup/legal-minds"
CLOUD_BUCKET="legal-minds-backup"
DATE=$(date +%Y%m%d)

# 使用 rclone 同步到云存储
rclone sync ${LOCAL_BACKUP_DIR} ${CLOUD_BUCKET} \
    --transfers=4 \
    --checkers=8 \
    --progress

# 清理云端 90 天前的备份
rclone delete ${CLOUD_BUCKET} \
    --min-age 90d \
    --progress

echo "[$(date)] 云同步完成" >> /var/log/legal-minds-backup.log
```

四、恢复流程

4.1 数据库恢复

```
#!/bin/bash
# restore_database.sh

BACKUP_FILE=$1
DB_PATH="/opt/legal-minds/data/legal_minds.db"

if [ -z "${BACKUP_FILE}" ]; then
    echo "用法: restore_database.sh <备份文件>"
    exit 1
fi

# 停止服务
systemctl stop legal-minds
```

```

# 备份当前数据库（以防万一）
if [ -f "${DB_PATH}" ]; then
    cp ${DB_PATH} ${DB_PATH}.backup.$(date +%Y%m%d_%H%M%S)
fi

# 解压备份
TEMP_DIR=$(mktemp -d)
tar -xzf ${BACKUP_FILE} -C ${TEMP_DIR}

# 恢复数据库
DB_FILE=$(find ${TEMP_DIR} -name "*.db" | head -1)
cp ${DB_FILE} ${DB_PATH}

# 设置权限
chown legal-minds:legal-minds ${DB_PATH}
chmod 640 ${DB_PATH}

# 清理
rm -rf ${TEMP_DIR}

# 启动服务
systemctl start legal-minds

# 验证
sqlite3 ${DB_PATH} "PRAGMA integrity_check;"
if [ $? -eq 0 ]; then
    echo "✅ 数据库恢复成功"
else
    echo "❌ 数据库恢复失败"
    # 回滚
    cp ${DB_PATH}.backup.* ${DB_PATH}
    exit 1
fi

```

4.2 文件恢复

```

#!/bin/bash
# restore_files.sh

```

```
BACKUP_FILE=$1
TARGET_DIR="/opt/legal-minds/storage"

if [ -z "${BACKUP_FILE}" ]; then
    echo "用法: restore_files.sh <备份文件>"
    exit 1
fi

# 解压恢复
echo "正在恢复文件..."
tar -xzf ${BACKUP_FILE} -C $(dirname ${TARGET_DIR})

# 设置权限
chown -R legal-minds:legal-minds ${TARGET_DIR}
chmod -R 750 ${TARGET_DIR}

echo "✅ 文件恢复完成"
```

4.3 完整恢复流程

```
#!/bin/bash
# full_restore.sh

DATABASE_BACKUP=$1
FILES_BACKUP=$2

echo "=== 开始完整恢复 ==="
echo "数据库备份: ${DATABASE_BACKUP}"
echo "文件备份: ${FILES_BACKUP}"

# 1. 停止服务
echo "1. 停止服务..."
systemctl stop legal-minds

# 2. 恢复数据库
echo "2. 恢复数据库..."
./restore_database.sh ${DATABASE_BACKUP}

if [ $? -ne 0 ]; then
    echo "❌ 数据库恢复失败"
    exit 1
fi
```

```
fi
```

```
# 3. 恢复文件
```

```
echo "3. 恢复文件..."
```

```
./restore_files.sh ${FILES_BACKUP}
```

```
# 4. 验证
```

```
echo "4. 验证恢复..."
```

```
sqlite3 /opt/legal-minds/data/legal_minds.db "SELECT COUNT(*) FROM cases;"  
CASE_COUNT=$?
```

```
if [ ${CASE_COUNT} -ge 0 ]; then
```

```
    echo "✅ 验证通过"
```

```
else
```

```
    echo "❌ 验证失败"
```

```
    exit 1
```

```
fi
```

```
# 5. 启动服务
```

```
echo "5. 启动服务..."
```

```
systemctl start legal-minds
```

```
# 6. 健康检查
```

```
echo "6. 健康检查..."
```

```
sleep 5
```

```
curl -f http://localhost:8000/api/v1/health
```

```
if [ $? -eq 0 ]; then
```

```
    echo "=== ✅ 完整恢复成功 ==="
```

```
else
```

```
    echo "=== ⚠️ 恢复完成但健康检查失败 ==="
```

```
fi
```

五、备份监控

5.1 备份检查脚本

```
#!/bin/bash
```

```
# check_backups.sh
```



```
BACKUP_DIR="/backup/legal-minds"
ALERT_EMAIL="admin@example.com"
MAX_AGE_DAYS=2
```

```
# 检查最新备份
```

```
LATEST_BACKUP=$(ls -t ${BACKUP_DIR}/database/*.tar.gz 2>/dev/null | head -
```

```
if [ -z "${LATEST_BACKUP}" ]; then
```

```
    echo "✗ 未找到备份文件"
```

```
    mail -s "备份告警：未找到备份" ${ALERT_EMAIL} < /dev/null
```

```
    exit 1
```

```
fi
```

```
# 检查备份时间
```

```
BACKUP_TIME=$(stat -c %Y ${LATEST_BACKUP})
```

```
CURRENT_TIME=$(date +%s)
```

```
AGE_HOURS=$(( (CURRENT_TIME - BACKUP_TIME) / 3600 ))
```

```
if [ ${AGE_HOURS} -gt $((MAX_AGE_DAYS * 24)) ]; then
```

```
    echo "✗ 备份超过${MAX_AGE_DAYS}天"
```

```
    mail -s "备份告警：备份过期" ${ALERT_EMAIL} < /dev/null
```

```
    exit 1
```

```
fi
```

```
# 检查备份大小
```

```
BACKUP_SIZE=$(du -m ${LATEST_BACKUP} | cut -f1)
```

```
if [ ${BACKUP_SIZE} -lt 10 ]; then
```

```
    echo "✗ 备份文件过小，可能异常"
```

```
    mail -s "备份告警：备份文件过小" ${ALERT_EMAIL} < /dev/null
```

```
    exit 1
```

```
fi
```

```
# 验证备份完整性
```

```
tar -tzf ${LATEST_BACKUP} > /dev/null 2>&1
```

```
if [ $? -ne 0 ]; then
```

```
    echo "✗ 备份文件损坏"
```

```
    mail -s "备份告警：备份文件损坏" ${ALERT_EMAIL} < /dev/null
```

```
    exit 1
```

```
fi
```

```
echo "✅ 备份检查通过"
echo "最新备份: ${LATEST_BACKUP}"
echo "备份时间: ${AGE_HOURS}小时前"
echo "备份大小: ${BACKUP_SIZE}MB"
```

5.2 定时任务

```
# crontab -e

# 每日备份 (凌晨 2 点)
0 2 * * * /opt/legal-minds/scripts/backup_database.sh >> /var/log/legal-mi

# 每周完全备份 (周日 1 点)
0 1 * * 0 /opt/legal-minds/scripts/backup_files.sh >> /var/log/legal-minds

# 每日备份检查 (早上 8 点)
0 8 * * * /opt/legal-minds/scripts/check_backups.sh >> /var/log/legal-mind

# 每月云同步 (1 号 3 点)
0 3 1 * * /opt/legal-minds/scripts/sync_to_cloud.sh >> /var/log/legal-mind
```

六、灾难恢复

6.1 灾难恢复计划

场景	恢复步骤	RTO	RPO
数据库损坏	从最近备份恢复	30 分钟	24 小时
文件丢失	从云备份恢复	1 小时	24 小时
服务器故障	新服务器 + 备份恢复	2 小时	24 小时
数据中心故障	切换到灾备中心	4 小时	24 小时

6.2 恢复时间目标

指标	目标	说明
RTO (恢复时间目标)	2 小时	从故障到恢复服务的时间
RPO (恢复点目标)	24 小时	最多丢失 24 小时数据

6.3 灾备演练

```
#!/bin/bash
# disaster_recovery_drill.sh

echo "=== 灾难恢复演练 ==="
echo "日期: $(date)"

# 1. 模拟数据库损坏
echo "1. 模拟数据库损坏..."
cp /opt/legal-minds/data/legal_minds.db /tmp/legal_minds.db.backup

# 2. 执行恢复
echo "2. 从备份恢复..."
LATEST_BACKUP=$(ls -t /backup/legal-minds/database/*.tar.gz | head -1)
/opt/legal-minds/scripts/restore_database.sh ${LATEST_BACKUP}

# 3. 验证数据
echo "3. 验证数据..."
CASE_COUNT=$(sqlite3 /opt/legal-minds/data/legal_minds.db "SELECT COUNT(*)")
echo "案件数量: ${CASE_COUNT}"

# 4. 恢复原状
echo "4. 恢复原状..."
cp /tmp/legal_minds.db.backup /opt/legal-minds/data/legal_minds.db

echo "=== 演练完成 ==="
```

七、备份清单

7.1 每日检查

- ☐ 备份任务执行成功
- ☐ 备份文件大小正常
- ☐ 备份日志无错误
- ☐ 云同步完成

7.2 每周检查

- ☐ 完全备份执行成功

- ☐ 备份恢复测试
- ☐ 备份存储空间检查

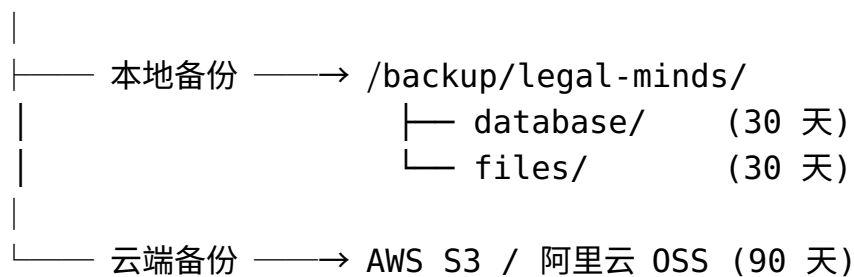
7.3 每月检查

- ☐ 灾难恢复演练
 - ☐ 备份策略审查
 - ☐ 备份保留策略清理
-

八、总结

8.1 备份架构图

生产服务器



8.2 恢复流程

故障发生



评估故障范围



选择恢复方案



停止服务



恢复数据



验证恢复



启动服务



健康检查

8.3 联系方式

角色	联系方式
----	------

技术负责人	王舟 15378391447
-------	----------------

运维负责人	[待填写]
-------	-------

紧急联系	[待填写]
------	-------

文档版本：v1.0

最后更新：2026 年 3 月 14 日

开发负责人：王舟

邮箱：main@mails.wedevs.org

电话：15378391447